

MVDream and The Consistent Multi-View Generation

Martina Cisotto

`martina.cisotto@studenti.unipd.it`

Abstract

MVDream is an innovative diffusion model designed to generate multi-view images from a text prompt. This model solves the Janus problem, a problem deeply connected with 2D-lifting methods, by generating consistent images from multiple viewpoints simultaneously. The role of multi-view consistency is extremely important to this approach, especially because it guarantees the integrity of an object in a virtual space. By analyzing MVDream, we highlight the advantages and limitations related to this method and the potential applications of it. We further briefly examine ImageDream, which incorporates image-prompting, and we underline the features in common with MVDream.

1. Introduction

The generation of 3D data is an important topic in our days. Although 2D-lifting methods have emerged, which are models that use pre-trained 2D diffusion models to supervise 3D creation, such as DreamFusion [1] and Magic3D [2], they often fail to maintain geometric consistency, leading to distorted output such as the Janus problem represented in Figure 1, where the model regenerates redundant features from different angles, and they can suffer from content drift, which refers to inconsistency across views, as we can see in Figure 2.

To solve this problem, multi-view diffusion models such as MVDream are proposed. In particular, MVDream generates simultaneously a set of multi-view images that are consistent with each other from a given text prompt. The concept of multi-view consistency is extremely important for MVDream because it leads to the maintenance of the integrity and complexity of an object from every perspective.

MVDream presents some advantages, such as the overcoming of Janus and content drift problem, the generation of complex assets from a simple textual description, and also some limitations such as the high necessity of memory and the low resolution.

We can also compare MVDream with ImageDream [3] by highlighting some similarities and differences. ImageDream is a 3D generative model that uses images as input

and generates different views of a particular object, such as MVDream.



Multi-face Janus Problem

Figure 1. The multi-view consistency problem: Janus Problem



Content Drift Problem

Figure 2. The multi-view consistency problem: Content Drift Problem

2. MVDream

The creation of 3D content is a fundamental but extremely hard process. For this reason, models which are able to generate 3D contents in an easy way are necessary. Currently, 3D generation methods are categorized into three main types: template-based generation pipeline, 3D generative models, and 2D-lifting methods. The first two methods have difficulties generalizing to arbitrary object generation, especially because of the limited accessible 3D models and large data complexity, and their generations are common and simple and have nothing in common with the complex and artistic structures requested in real-world.

2D-lifting methods use pre-trained 2D diffusion models to supervise the optimization of 3D representations. However, these approaches face critical hurdles due to a lack of multi-view knowledge. The most significant issue is the multi-face Janus problem, where the system frequently regenerates redundant features from different angles. In addition, a content drift occurs, leading to a loss of identity as the viewpoint changes.

In the paper "MVDream: Multi-view diffusion for 3D generation" [4], MVDream is analyzed, which is a multi-

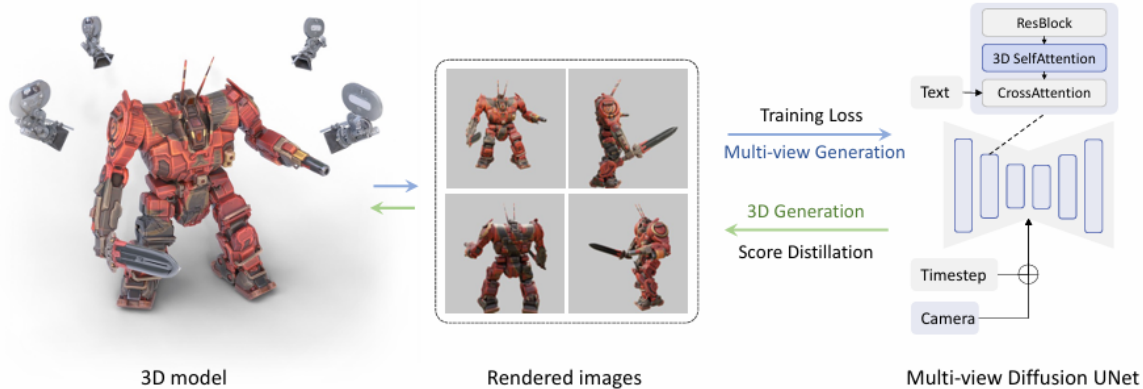


Figure 3. The multi-view diffusion model: multi-view renderings are used to train the model and, for testing, the 3D representation is optimized via Score Distillation Sampling (SDS)

view diffusion model designed to generate consistent multi-view images from a given text prompt. MVDream achieves the generalizability of 2D diffusion models while maintaining the consistency of 3D renderings. This approach guarantees stability in 3D generation and, if we also consider the DreamBooth approach[5], the ability to learn new concepts from a collection of 2D images. MVDream learns

$$p(I_1, I_2, \dots, I_n | \text{text}, C_1, C_2, \dots, C_n)$$

where each I_i points to an absolute camera pose C_i , therefore it generates all views together in only one diffusion process. In this way, it processes multiple latent images simultaneously, where each latent corresponds to a camera view, and the latents interact through cross-view attention. This method allows the model to maintain the consistent structure and shape of the object. By informing the model how the object should appear from every angle, it can learn multi-view correlations and 3D structure. (see Figure 3)

3. Multi-view consistency

The concept of multi-view consistency serves as the structural backbone of MVDream, directly addressing the core limitations of traditional 2D-lifting methods. Unlike standard models that interpret 3D objects as a series of independent 2D snapshots, a multi-view approach treats the object as a unified spatial entity.

Considering 2D-lifting methods, a way to improve their viewpoint-awareness can be adding viewpoint descriptions to text conditions, or another way can be considering video diffusion models, but if there is a large viewpoint change there can be the content drift problem.

By directly training a multi-view diffusion model, we obtain a set of images of the same scene from different perspectives. Synchronized exchange of information ensures that features such as geometry, texturing, and lighting remain stable and coherent as the viewpoint changes, eliminating

content drift, and maintaining the integrity of an object from every perspective.

4. Advantages and limitations of MVDream

To understand the potential of MVDream, it is necessary to examine the advantages and limitations of this method. First of all, by using a multi-view approach, the model ensures that all generated views are geometrically connected and consistent with each other during the diffusion process, effectively solving the Janus Problem and content drift common in 2D-lifting methods. MVDream inherits the knowledge and the creative capabilities of 2D diffusion model, such as Stable Diffusion, so it can generate artistic and complex assets from simple text prompts while maintaining 3D awareness by training on a mixture of 2D and 3D datasets, and it is more stable than single-view 2D diffusion models. Furthermore, these methods can generate 3D content in an easy way producing high-quality assets, so it can also be used by non professional users, saving a huge amount of time and costs. Finally, speaking of Multi-View Dreambooth [5], it can learn a specific new concept from just a few 2D examples and generate them in 3D with robust consistency.

On the other hand, the model generates only images, so it cannot be considered a real 3D model, and it trains and optimizes at a lower resolution (256x256) respecting the original stable diffusion, that is, 512x512, and for this reason it might limit the ultra-fine surface details compared to high-resolution 2D generators. In addition, generating multiple views simultaneously increases the memory and computational requirements. Another limitation is that the model depends on the dataset, in particular on the quality of the multi-view data; in fact to improve quality, the model requires more diverse and realistic 3D renderings, which is noted as being potentially costing. The visual style of the generated objects, including lighting and texture, is in-

fluenced by the specific 3D rendered datasets used during training, this can lead to a lack of the full realism of natural images.

5. Potential applications of MVDream

The technical solutions related to MVDream, such as the attention of the details and the multi-view consistency, are important topics for some real-world applications. A significant application for MVDream lies in personalized and automated 3D asset pipelines for independent game developers. Creating a unique 3D character requires hours of professional modeling and high-level 3D modeling skills. Using the Multi-View DreamBooth capability [5], a developer could provide a few sketches or reference images of their unique character. MVDream would assimilate this identity and generate consistent multi-view images that serve as a prior for creating a high-fidelity 3D model. This application requires the model to maintain complex and unique character details, and MVDream ensures that the resulting 3D-printed character will not have multiple faces or distorted features (because MVDream solves the Janus problem). Therefore, people can create personalized characters and assets that maintain their structural integrity from every angle.

In the context of e-commerce, the MVDream approach can be utilized to revolutionize the visualization of virtual products. By its ability to generate consistent multi-view images from text prompts, the model can lead to the creation of high-quality digital twins of products without the need for labor-intensive manual modeling. MVDream ensures that a potential buyer can rotate a virtual item and see a geometrically accurate and stable representation from any angle. This level of 3D awareness reduces the uncertainty of online shopping, providing a true virtual experience.

6. MVDream and ImageDream: similarities and differences

Image Dream [3] is an innovative 3D generative model that uses an image as input that can capture precise textures and colors, and the exact position in the space related to other objects. Both MVDream and ImageDream are multi-view diffusion models (Figure 4): they do not generate a single image at one time but they generate simultaneously different views of an object that are coherent with each other. They create 3D models with a correct geometry and high-quality texture to overtake the limitations connected to 2D-lifting methods such as the Janus problem and the content drift problem. ImageDream introduces more granular control through a multi-level controller that integrates global, local, and pixel-level information directly from the input image.

The main difference between these two approaches is

that the input of MVDream is a text, instead the input of ImageDream is an image, in this way it guaranties the presence of more precise and particular details. ImageDream obtains a better image-text alignment, the text stabilizes what is generated, while the image stabilizes how the object exactly must appear, so ImageDream generates extremely correct geometries from an input-image. On the other hand, MVDream has more creative freedom, is more fleasible but has less control on details because its input is a textual description. Ultimately, ImageDream surpasses MVDream in its ability to preserve visual details that are difficult to capture through text alone, combining the stability of MVDream with the rich informational depth of image conditioning.

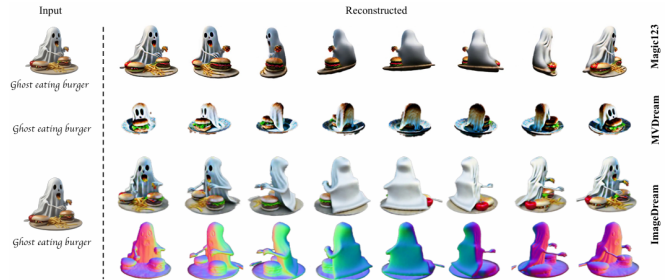


Figure 4. Comparison between MVDream and ImageDream output

7. Conclusions

MVDream is a model which is able to generate simultaneously a set of multi-view images from a text-input. We have seen that this model is capable of solving the Janus problem and content drift, but it is extremely connected with the quality of the considered data, and an improvement in quality might potentially cost. This approach can be very helpful in some real world applications such as the gaming industry, especially for creating 3D-printend characters with their structural integrity for every angle. We have also seen that ImageDream has better results with respect to MVDream because of the utilize of an input-image which stabilizes how exactly an object must appear.

References

- [1] Ben Poole Ajay Jain Jonathan T. Barron and Ben Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. *In ICLR*, 2023.
- [2] Chen-Hsuan Lin Jun Gao Luming Tang Towaki Takikawa Xiao-hui Zeng Xun Huang Karsten Kreis Sanja Fidler Ming-Yu Liu and Tsung-Yi Lin. Magic3d: High-resolution text-to-3d content creation. *In CVPR*, 2023.
- [3] Peng Wang Yichun Shi. Imagedream: Image-prompt multi-view diffusion for 3d generation. *arXiv:2312.02201v1*, 2023.
- [4] Yichun Shi Peng Wang Jianglong Ye Long Mai Kejie Li Xiao Yang. Mvdream: Multi-view diffusion for 3d generation. *arXiv:2308.16512v4*, 2024.

- [5] Nataniel Ruiz Yuanzhen Li Varun Jampani Yael Pritch Michael Rubinstein and Kfir Aberman. Dream-booth: Fine tuning text-to-image diffusion models for subject-driven generation. *In CVPR*, 2023.